

TECHNICAL INCIDENT INVESTIGATION REPORT

FORENSIC DETECTION, SOURCE ATTRIBUTION & COASTAL THREAT ASSESSMENT

CLASSIFICATION NOTICE: Contains analytical assessments derived from synthetic/demo SAR telemetry, Euler hindcast particle integration, and AIS track correlation. Candidate vessel ratings denote statistical association and do NOT constitute legal proof of culpability.

DETECTION SUMMARY		SIMULATED
Spill Detected	True	
Num Detections	1	
Total Detections	2	
Validation Status	CONFIRMED BY MULTIPLE SIGNALS	
Detections	[{'detection_id': 1, 'confidence': 0.4823, 'polygon_points': 189, 'bbox': [187, ...	
All Detections	[{'detection_id': 0, 'confidence': 0.6216, 'polygon_points': 4, 'bbox': [2, 3, 5...	
Image Shape	[512, 512, 3]	
Model Name	best.pt	
Inference Time Ms	4922.30	
Source File	C:\Coding\Jal-Rakshak\demo\demo_sar_patch.png	

SPILL CHARACTERIZATION		SIMULATED
Centroid Px	[267.6, 241.1]	
Perimeter Px	517.50	
Area Px	17725.50	
Length Px	180.30	
Width Px	125.60	
Aspect Ratio	1.44	
Orientation Deg	124.90	
Num Connected Regions	1	
Is Fragmented	False	
Convexity	0.9850	
Solidity	0.9850	
Pixel Resolution M	10.00	
Area Sq Km	1.77	
Perimeter Km	5.17	
Length Km	1.80	
Width Km	1.26	
Volume Estimation Method	empirical	
Estimated Volume Tons	753.33	
Volume Note	Estimated using empirical thickness assumption (0.1-1.0 mm). This is NOT a measured quantity. Actual volume depends on oil type, weathering, and spill dynamics.	
Centroid Lat	12.45	
Centroid Lon	80.23	

SPILL AGE & WEATHERING DYNAMICS		INFERRED
Status	ESTIMATED	
Estimated Age Range Hours	[0.5, 2.5]	
Best Estimate Hours	1.20	
Confidence	0.6500	
Evidence Method	fay_morphology_heuristic	
Supporting Evidence	['Observed aspect ratio: 1.44 (Major axis 1.80 km / Minor axis 1.26 km).', 'Obse...	
Uncertainty Factors	['Single-scene estimation: no prior satellite pass available to verify release s...	
Fay Regime	gravity_viscous	

DETECTION VALIDATION		INFERRED
Yolo Confidence	0.4823	
Classical Agreement	0.6041	
Look Alike Risk	0.3500	
Land Sea Consistency	1.00	
Morphology Consistency	0.6000	
Contrast Ratio	0.3170	
Final Validation Status	CONFIRMED BY MULTIPLE SIGNALS	
Overlap Score	0.2082	
Explanation	Confirmed: Strong multi-signal agreement (YOLO 48%, classical agreement 60%, dark damping ratio 0.32).	
Method Results	[{'method': 'adaptive_threshold', 'num_regions': 982, 'total_area_px': 86487.0, ...}]	

SOURCE RECONSTRUCTION (HINDCAST)		PREDICTED
Origin Lat	12.46	
Origin Lon	80.21	
Origin Time	2026-09-14T14:20:00+00:00	
Origin Time Str	14:20 UTC	
Destination Lat	12.45	
Destination Lon	80.23	
Destination Time	2026-09-14T15:30:00+00:00	
Total Distance Km	2.97	
Temporal Uncertainty Min	36.00	
Spatial Uncertainty Km	1.00	
Current Speed Ms	0.4800	
Current Bearing Deg	118.00	
Wind Speed Ms	6.20	
Wind Bearing Deg	135.00	
Method	euler	
Direction	hindcast	
Num Trajectory Points	15	

PROBABLE SOURCE LIKELIHOOD SURFACE		INFERRED
Peak Location	12.4631, 80.2092	
Dispersion Sigma Km	0.5000	
P50 Radius Km	0.5900	
P95 Radius Km	1.22	
Core Area Sq Km	1.09	
Uncertainty Statement	Probable source region modeled with Gaussian dispersion $\sigma = 0.5$ km. Core 50% confidence area is 1.1 km²; 95% boundary extends 1.2 km from peak. Surface accounts for current shear and turbulent diffusion. Does NOT represent exact physical boundaries.	

CANDIDATE VESSEL ANALYSIS	INFERRED
No candidate vessels correlated.	

DRIFT PREDICTION (FORECAST)	PREDICTED
Forecasts	[{'origin_lat': 12.45, 'origin_lon': 80.23, 'origin_time': '2026-09-14T15:30:00+...

COASTAL & ENVIRONMENTAL IMPACT ASSESSMENT	PREDICTED
Nearest Shoreline	Kalpakkam
Shortest Distance Km	13.86
Landfall Projected	False
Eta To Coast Hours	None
Eta Uncertainty Range Hours	None
Coastal Vulnerability Score	38.80
Threatened Assets Count	1
Threatened Assets	[{'name': 'Madras Atomic Power Station (MAPS) Seawater Intake', 'category': 'inf...
Containment Recommendations	['Deploy ocean containment booms at earliest daylight or calm sea state', 'Notif...
Dispersant Restrictions	['CRITICAL: Chemical dispersants prohibited within 10 km of shoreline (NEPA/CPCB...

RISK ASSESSMENT

INFERRED

Level	MEDIUM
Overall Score	36.30
Factors	[{'name': 'Spill Area', 'value': 1.77255, 'score': 60, 'weight': 0.2, 'descripti...
Recommendations	['Protect sensitive areas with boom deployment']
Alert Required	True
Alert Priority	standard

ALERTS GENERATED

OBSERVED

Incident Id	JR-20260928-183114
Total Alerts	2
Alerts	[{'alert_id': 'COMM-20260928183114', 'timestamp': '2026-09-28T18:31:14.209892+00...

METHODOLOGICAL LIMITATIONS & UNCERTAINTIES

- Spill age estimation uses morphological elongation and Fay spreading theory. Without multi-temporal satellite passes, age bounds represent empirical approximations.
- Hindcast trajectory assumes constant current/wind fields. Real conditions may vary significantly over the drift period.
- Vessel attribution is probabilistic and based on available AIS data. AIS data gaps, spoofing, or incomplete coverage may affect results.
- Shoreline landfall ETA depends on hydrodynamic models and wind leeway factor (0.03). Tidal flux and nearshore surf zone dynamics may modify actual arrival time.
- This report is generated by an AI system for decision support. It does not independently establish legal liability.
- This report was generated using SIMULATED/DEMO data. All values are synthetic and for demonstration purposes only.

INTEGRATED DATA PROVIDERS & MODELS

Primary Feeds: YOLO v8 Segmentation Model, Classical SAR Analysis, Demo/Simulated Data

SYSTEM ANALYST: Automated Algorithmic Dispatch	REVIEWING OFFICER: _____	COMMAND ACTION: [] DISPATCH INVESTIGATION [] ESCALATE ALERT [] ARCHIVE RECORD
---	-----------------------------	---